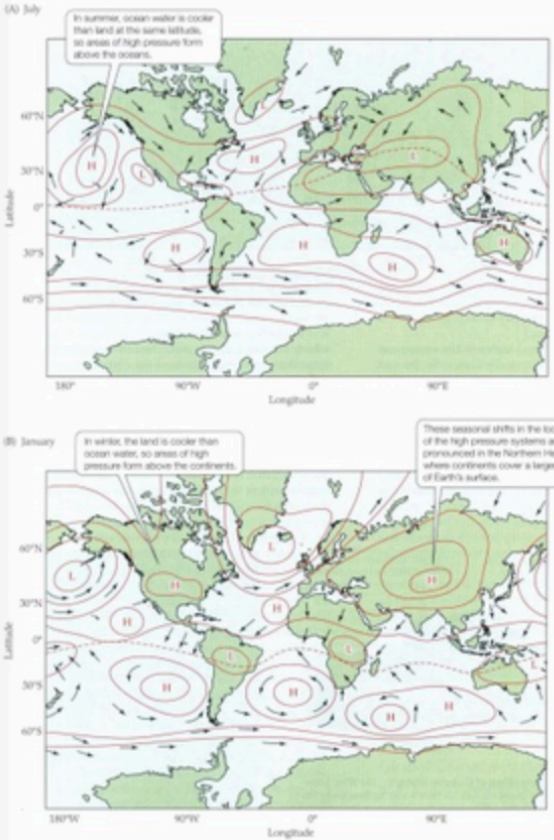


Figure 2.10 Prevailing Wind Patterns The difference in heat capacity between the oceans and the continents leads to seasonal changes in atmospheric pressure cells that influence prevailing wind patterns.

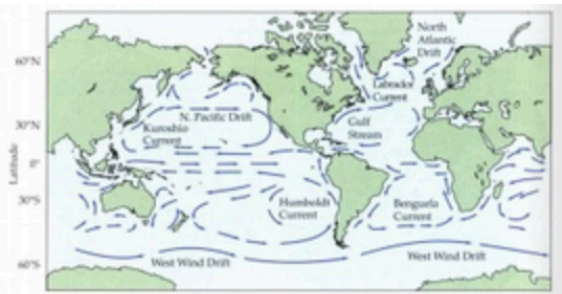


Figure 2.11 Global Ocean Surface Currents The major ocean surface currents are driven by the surface winds shown in Figure 2.10 but modified by the Coriolis effect.

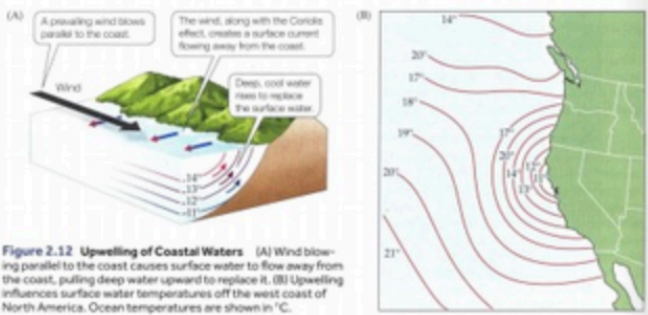


Figure 2.12 Upwelling of Coastal Waters (A) Wind blowing parallel to the coast causes surface water to flow away from the coast, pulling deep water upward to replace it. (B) Upwelling influences surface water temperatures off the west coast of North America. Ocean temperatures are shown in °C.

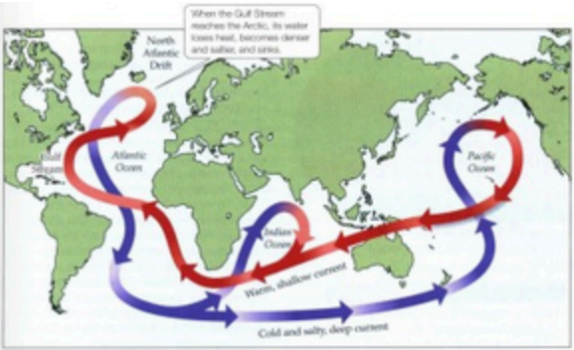


Figure 2.13 The Great Ocean Conveyor Belt
An interconnected system of surface and deep ocean currents transfers energy between tropical and polar regions.